



Effective and compact multimodal molecular representation optimization with molecular fragments enhancement

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ABSTRACT

Molecular representation learning has attracted considerable attention in recent years, particularly in the field of AI-driven drug discovery. Many existing studies optimize models and learn high-quality molecular representations efficiently by exploiting multimodal information for self-supervised learning. However, these methods often overlook critical functional group fragments within molecules and fail to address redundancy and sufficiency among molecular modalities, leading to suboptimal performance. To address these concerns, we propose a novel framework called Effective and Compact Multimodal Molecular Representation Optimization with Molecular Fragments Enhancement (ECMMR). Specifically, we first leverage the BRICS algorithm to decompose molecules into molecular fragments containing functional groups, and construct molecular hypergraphs based on these fragments to enrich the informative representations. We then design contrastive and generative self-supervised tasks to learn modality-consistent information and learn to optimize cross-modal latent representations from a coupled-priori optimization perspective. This enables mutual reinforcement between 2D and 3D graph encoders while reducing redundancy. Extensive experiments demonstrate that our approach achieves superior performance on 22 downstream tasks.

1. Introduction

Recent advancements in machine learning have captured significant attention from biochemical informatics researchers, particularly in the area of molecular representation learning (MRL) (Deng et al., 2023; Guo et al., 2023; Li et al., 2022; Lin et al., 2024; Liu et al., 2024; Zhu et al., 2023). Effective molecular representation forms the foundation for the challenging drug discovery problems (Hu et al., 2019; Zaidi et al., 2023). These models can accurately capture the complex structure and chemical properties of molecules, thereby improving the efficiency and success rate of drug discovery.

Molecules can be represented in diverse formats, such as the Simplified Molecular Input Line Entry System (SMILES), molecular graphs, and

others. Each representation highlights unique molecular characteristics, such as atomic composition, topology, and spatial arrangement. However, single-modality methods (Chithrananda et al., 2020; Wen et al., 2022) often fail to capture the complexity of molecular properties, resulting in limited predictive performance. It is therefore intuitive that multimodal MRL, which integrates information from multiple perspectives, has emerged as a promising solution (Chen et al., 2025; Liu et al., 2023, 2021; Stärk et al., 2022; Yi et al., 2025; Zheng et al., 2024, 2023). Specifically, numerous self-supervised methods that benefit from well-designed agent tasks have been developed. For example, Stärk et al. proposed maximizing the mutual information between 3D summary vectors and graph neural network representations (Stärk et al., 2022), while Liu et al. introduced mutual contrast and reconstruction of molecular graph

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representations (Liu et al., 2021). Both approaches enhance the consistency of molecular modalities, but they do not fully address the issues of redundancy and sufficiency between modalities, which may lead to representation inefficiency.

Furthermore, while these self-supervised methods offer an efficient approach to learning representation, they often prioritize global features of the molecule, overlooking the significance of local structures derived from chemical knowledge. It is well known that molecular fragments with functional groups are the key structural units in a molecule, and they have a significant impact on the physicochemical properties of a molecule. For example, the hydroxyl group -OH plays a key role in determining a molecule's water solubility. Recently, several methods have been proposed that leverage functional groups or fragments. For example, Zhang et al. improved the Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) (Degen et al., 2008) algorithm to partition molecules and proposed a generative pre-training framework for generating molecular graphs step-by-step (Zhang et al., 2021). Chen et al. proposed molecular hypergraph neural network to represent higher-order conjugate structures as hyper edges to capture information about subgraph structures (Chen & Schwaller, 2024). While both approaches effectively learn subgraphs to capture rich structural and semantic information, they operate independently of the 2D modality. As a result, the complementary information inherent in molecular multimodality remains underutilized.

Effective molecular representation is crucial for downstream tasks. From the above analysis of multimodal MRL, we can decompose these challenges into two questions: (1) Lack of emphasis on functional group fragments within the molecule. Existing multimodal approaches predominantly focus on aligning overall molecular properties while neglecting functional group fragments with distinct chemical significance. These fragments are pivotal in determining a molecule's physicochemical properties and directly impact the accuracy of drug discovery. The impact of key fragments for downstream tasks cannot be ignored. (2) Redundant information between modalities. While current contrastive learning or simple generation tasks enhance modality consistency, they fail to address redundancy between 2D and 3D information. This results in inefficient representations and limits the ability to fully exploit complementary information across modalities. Redundant information can prevent the model from fully utilizing the complementary data provided by different molecular modalities. Conversely, a lack of sufficiency indicates that some modalities fail to provide adequate information to effectively support downstream tasks. These challenges highlight the need for improved multimodal MRL, specifically in fully leveraging molecular inner structure information and effectively integrating multi-modal data.

To address these concerns, we propose Effective and Compact Multimodal Molecular Representation Optimization with Molecular Fragments Enhancement (ECMMR). The overall framework of ECMMR is illustrated in Fig. 1. Our main objective is to effectively leverage the inner structure information of molecule and integrate multimodal data. Specifically, we first decompose the molecule into fragments containing functional groups using the BRICS algorithm, which incorporates prior knowledge of chemical reactions. To better capture information from functional group fragments, we construct hypergraphs derived from these molecular fragments. The hypergraph's key feature is the hyperedges, which can connect multiple nodes, allowing to capture the many-to-many relationships within complex molecular fragments. This approach enables better modeling of the intricate structure within molecule. Building on this, we develop a self-supervised learning framework designed to learn the intrinsic structures and patterns of molecules. We design contrastive and generative self-supervised proxy tasks to allow molecular 2D and 3D encoders to learn complementary information. For the contrastive task, we perform contrastive learning at the node level to estimate the mutual information between the two modalities. For the generative proxy task, molecular 2D and 3D modalities are reconstructed from each other in the latent feature space. We aim

to achieve consistency of the different modalities in the latent feature space, facilitating the transfer of reliable semantic information between them. To further optimize the learned molecular representations, we optimize the generation task from the perspective of coupled prior optimization. We propose to optimize a sufficiently variational representation lower bound to enhance consistency between modalities and reduce redundant information to learn more sufficient representations.

The main contributions of our work are as follows:

- Distinct from the conventional approaches that treat molecules as simple atomic graphs, we propose leveraging BRICS-based chemical reaction prior knowledge to guide molecular fragment extraction. By constructing molecular hypergraphs based on these fragments, our method explicitly captures the many-to-many higher-order interactions within and between functional groups-interactions often overlooked by standard GNNs but critical for determining molecular properties.
- To address the widespread issues of information redundancy and insufficiency in existing molecular 2D and 3D alignment methods, we propose a dual-task framework comprising node-level contrastive learning and a sufficient generation proxy task. Unlike simple global alignment, our framework minimizes redundant noise from a coupled prior optimization perspective while ensuring bidirectional transmission of reliable semantic information, thereby achieving more compact and information-rich latent representations.
- Extensive experiments verified the effectiveness of ECMMR, which shows superior performance on 22 downstream tasks.

2. Related work

2.1. Molecular representation learning

There is extensive research in Molecular Representation Learning for different modality or views of molecules. Hu et al. designed mask-based self-supervised tasks on molecular 2D graphs for atoms or bonds (Hu et al., 2019). Sun et al. proposed to maximize the mutual information of molecular graph level features and substructure features (Sun et al., 2020). Wang et al. proposed a self-supervised learning framework based on contrastive learning with data augmentation on molecular graphs (Wang et al., 2022). Zhang et al. optimize the molecular fragmentation method and propose a self-supervised framework based on motif generation (Zhang et al., 2021). In recent years, multi-modal based methods have received much attention to exploit the complementary information of molecule. Stärk et al. introduced 3D molecular structure information with 2D molecular graph representation to construct pre-training based on contrastive learning (Stärk et al., 2022). Liu et al. enhanced the capability of molecular encoder by fusing two agent tasks, contrastive and generative learning (Liu et al., 2021). Further, Liu et al. proposed to improve the generative task using group symmetric stochastic differential equation models on the data space, which significantly enhanced the modeling capability of the molecular encoder (Liu et al., 2023). Chen et al. proposed learning molecular representations by fusing multimodal information from 2D topological and 3D geometric modalities, utilizing self-supervised tasks at both node-level and graph-level (Chen et al., 2025). Existing research has focused on global-level modal alignment. We propose utilizing BRICS-based fragment decomposition to explicitly capture fine-grained chemical semantics within functional groups. Unlike standard graph-based approaches or previous motif-based SSL methods, we introduce hypergraph modeling to represent higher-order relationships between atoms and fragments, effectively encapsulating molecular structural hierarchies. Furthermore, existing methods overlook cross-modal redundancy issues. We propose shifting the paradigm from simple feature fusion to strategy refinement, filtering redundant noise while preserving core information.

2.2. Hypergraph

Hypergraphs can model complex relational networks. An hyper-edge in a hypergraph can connect any number of nodes, which makes hypergraphs more flexible to express many-to-many relationships. Feng et al. proposed hypergraph neural network to learn data representation considering complex correlations of data (Feng et al., 2019). Chen et al. represented conjugate structures as hyperedges and proposed molecular hypergraph neural network (Chen & Schwaller, 2024). Although these methods demonstrate the flexibility of combining hypergraphs with neural networks, hypergraph modeling in our framework serves to integrate chemical fragments into multimodal learning. The rich information contained within these hypergraphs provides a more comprehensive foundation for subsequent sufficiency optimization.

2.3. Information bottleneck

Information Bottleneck (IB) is a theoretical framework for understanding and optimizing information processing in the learning process. IB theory emphasizes optimizing the representation so that it contains enough information to accomplish a specific task without containing too much noise or irrelevant details, as a way to improve the efficiency of the learning process and the predictive performance of the model. Tishby et al. (2000). IB theory is widely used in several fields (Cui et al., 2024; Federici et al., 2020; Li et al., 2023; Sun et al., 2022). Its wide application in several fields highlights the practical value of information bottleneck theory. In ECMMR, IB theory provides the optimization principle. This selective refinement ensures that the final representation not only effectively captures chemical subtleties but also remains compact and robust against noise.

3. Methodology

3.1. Problem definition

We denote the molecular graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X, R)$, where $\mathcal{V} = \{v_i\}_{i=1}^N$ denotes the set of N atomic nodes, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents chemical bonds with adjacency matrix $A \in \{0, 1\}^{N \times N}$ where $A_{ij} = 1$ if $(v_i, v_j) \in \mathcal{E}$, $X \in \mathbb{R}^{N \times d_a}$ contains d_a -dimensional node attributes, $E \in \mathbb{R}^{|\mathcal{E}| \times d_e}$ denotes edge features for existing edges, and $R \in \mathbb{R}^{N \times 3}$ is the 3D coordinate matrix of the atoms. We consider a combination of generative and contrastive tasks for pre-training to learn from large-scale unlabeled datasets to develop robust molecular representations.

3.2. Molecular fragment-enhanced hypergraph

To enhance the encoder's ability to learn from molecular fragments with functional groups, we introduce hypergraphs to expand the modeling capabilities of traditional Graph Neural Network (GNN). In conventional graph structures, the focus is primarily on binary relations, where each edge connects only two nodes. However, when molecular fragments consist of multiple atoms, handling complex many-to-many relationships and structures directly becomes challenging. To address this, we establish hyperedges that connect these fragments with their constituent atoms, enabling feature transformation across groups of atoms within the hyperedges. This approach allows the GNN to more effectively capture and learn the rich information embedded in molecular fragments.

Molecular 2D and 3D graph representation. Firstly, we encode the molecular 2D topological graph by Graph Isomorphism Network (GIN) (Xu et al., 2018), leveraging its provable expressiveness in distinguishing graph isomorphism classes. Given a molecular graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node features X and edge features E , the node-wise update at layer l is

defined as:

$$Z_i^{2D,(l+1)} = \text{MLP}_{node}^{l+1} \left((1 + \epsilon^{(l)}) \cdot z_i^{2D,(l)} + \sum_{j \in \mathcal{N}(i)} \left[z_j^{2D,(l)} + \text{MLP}_{edge}^{(l+1)}(e_{ij}) \right] \right), \quad (1)$$

where $Z_i^{2D,(l)}$ is the input feature of node i at layer l , $Z_i^{2D,(0)} = x_i$, $\mathcal{N}(i)$ is the neighborhood set of atom i , and $\epsilon^{(l)}$ is a learnable scalar parameter. MLP_{node} and MLP_{edge} denote MultiLayer Perceptrons (MLPs) applied to nodes and edges, respectively. The 2D atom representation is the output of the last layer, and mean pooling is further employed to obtain the 2D molecular graph representation h_{2D} .

For the molecular 3D space, modeling the coordinates and relative distances of the atoms helps the model to understand the molecule. Similar to most previous methods, we adopt (Schütt et al., 2017) continuous-filter convolutional layers that learn distance-aware interactions. For atom i with coordinates r_i , the 3D representation update at layer k is:

$$Z_i^{3D,(k+1)} = \sigma \left(W_{self}^{(k)} z_i^{3D,(k)} + \sum_{j \neq i} \phi(\|r_i - r_j\|_2) \cdot W_{inter}^{(k)} z_j^{3D,(k)} \right), \quad (2)$$

where $\phi: \mathbb{R} \rightarrow \mathbb{R}^d$ is a radial basis function (RBF) projection: $\phi(r) = \exp(-\gamma \|r - \mu\|_2^2)$, with learnable parameters γ , μ and $\sigma(\cdot)$ denotes the Shifted Softplus activation. The 3D molecular representation is obtained via geometric mean pooling.

Molecular fragmentation and hypergraph construction. We use the BRICS algorithm to fragment molecules. BRICS analyzes molecular structures based on chemical reaction perspectives, decomposing molecules into fragments that contain specific functional groups.

Definition 3.1 (Molecular Fragmentation). Given a molecule M , the BRICS algorithm decomposes M into a set of non-overlapping fragments $F_M = \{f_1, f_2, \dots, f_m\}$, where each fragment f_i corresponds to a functional group or pharmacophore. These fragments satisfy the completeness $\bigcup_{i=1}^m f_i = M$ and non-redundancy $f_i \cap f_j = \emptyset$ for $i \neq j$.

Definition 3.2 (Molecular Hypergraph). The hypergraph representation of M is a triple $\mathcal{H}_M = (\mathcal{V}, \mathcal{E}_H, X)$, where $\mathcal{V} = [v_1, v_2, \dots, v_{|\mathcal{V}|}]$ is the set of atomic nodes and $\mathcal{E}_H = [e_1, e_2, \dots, e_{|\mathcal{E}_H|}]$ is the set of hyperedges. The incidence matrix $H \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{E}|}$ is defined as:

$$h_{ve} = \begin{cases} 1, & \text{if } v \in e \\ 0, & \text{if } v \notin e \end{cases}. \quad (3)$$

Proposition 1 (Non-overlapping Hyperedges). Since BRICS fragments are non-redundant, the hyperedges in $\mathcal{H}_M$ form a partition of $\mathcal{V}$. Formally:

$$\forall v \in \mathcal{V}, \exists! e \in \mathcal{E}_H \text{ such that } v \in e. \quad (4)$$

Proof. Directly follows from Definition 1. Each atom belongs to exactly one fragment, hence one hyperedge.

Hyperedge features are obtained by specific operations based on node features. Node features Z_δ are projected into a latent space via a learnable weight matrix $\theta \in \mathbb{R}^{d \times d_h}$. Features of hyperedge e are computed by aggregating the features of its constituent nodes using H^T . Hyperedge features are transformed via a linear layer $a_\psi \in \mathbb{R}^{d_h \times d_h}$ and propagated back to nodes. The final node features combine the propagated hyperedge features and original node features. This process can be formulated as:

$$Z_\delta = a_\psi(HW a_\phi(H^T Z_\delta \theta)) + Z_\delta, \quad \delta \in [2D, 3D], \quad (5)$$

where a_ϕ is linear layers.

To balance local atomic features with global fragment information, we choose to construct molecular fragment-enhanced hypergraphs only in the final layer of the GNN. This design prevents excessive global noise interference during the initial feature extraction stage, ensuring the model first captures robust atomic features at lower levels before aggregating higher-level functional group semantics through hypergraphs at the top layer. This achieves effective information integration from local to global scales.

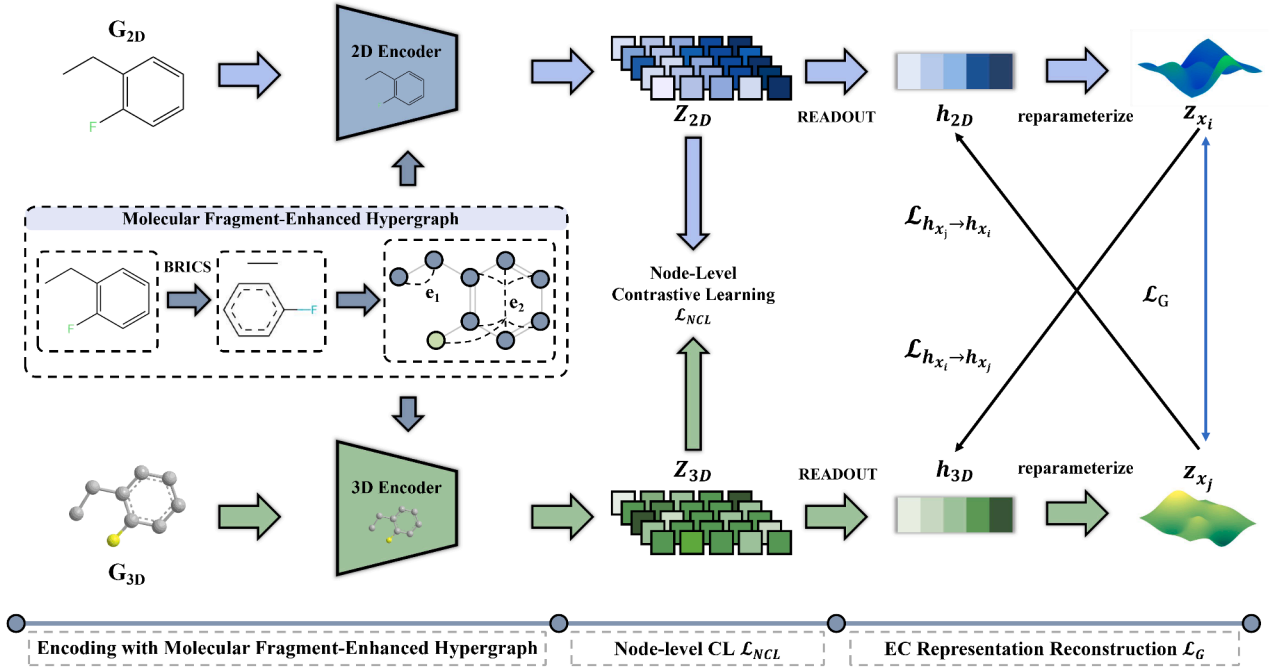


Fig. 1. Illustration of the proposed ECMMR framework. (1) Encoding with Molecular Fragment-Enhanced Hypergraph: 2D and 3D molecular maps are encoded by their respective encoders and the process is further optimized by Molecular Fragment-Enhanced Hypergraph (Section 3.2). (2) Effective and Compact Representation Reconstruction: We propose a generative agent task to optimize molecular representations in latent space (Section 3.3.2). (3) Node-Level CL: We introduce a contrastive proxy task to jointly optimize with the generative proxy (Section 3.3.3).

3.3. ECMMR self-supervised framework

In the ECMMR framework, we design contrastive and generative self-supervised proxy tasks that aim to enable the molecular 2D encoder and 3D encoder to learn complementary information. In this way, the two views are facilitated to achieve a bidirectional injection of knowledge between the 2D and 3D representations. Firstly, we start with the molecular multi-modal mutual information perspective. Then, we introduce the contrastive and generative tasks in ECMMR, respectively. Finally, the training objective is given.

3.3.1. Cross-modal mutual information

Maximizing mutual information (MI) is a commonly applied technique for learning representations in multi-modal and multi-view learning. The molecular self-supervised approach can be achieved by maximizing the MI between modalities. The formalization is as follows:

$$\mathcal{L}_{MI} = \frac{1}{2} \mathbb{E}_{p(x,y)} [\log p(x|y) + \log p(y|x)], \quad (6)$$

where x and y represent different modalities respectively in here.

Definition B.1 (Mutual information). For stochastic systems X and Y , the mutual information is defined as the Kullback-Leibler divergence of the joint distribution $p(x, y)$ and the product of the marginal distributions $p(x)p(y)$:

$$I(X; Y) = D_{KL}(p(x, y) \| p(x)p(y)) = \mathbb{E}_{p(x,y)} \left[\log \frac{p(x, y)}{p(x)p(y)} \right]. \quad (7)$$

Lemma B.2 (Symmetric Conditional Entropy Reconstruction). Mutual information can be decomposed into a combination of symmetric conditional entropy and correction divergence:

$$I(X; Y) = \frac{1}{2} [H(Y|X) + H(X|Y)] + \frac{1}{2} D_{KL}(p(x, y) \| \sqrt{p(x|y)p(y|x)}), \quad (8)$$

where the conditional entropy is defined as $H(Y|X) = -\mathbb{E}_{p(x,y)} [\log p(y|x)]$.

Proof. Expand the mutual information definition and reorganize the terms:

$$\begin{aligned} I(X; Y) &= \mathbb{E}_{p(x,y)} \log \frac{p(x, y)}{p(x)p(y)} \\ &= \mathbb{E}_{p(x,y)} \log \frac{p(x, y)}{\sqrt{p(x)p(y)} \cdot \sqrt{p(x)p(y)}} \\ &= \frac{1}{2} \mathbb{E}_{p(x,y)} \log \frac{p(x, y)^2}{p(x)p(y)p(x)p(y)} \\ &= \frac{1}{2} \mathbb{E}_{p(x,y)} [\log p(y|x) + \log p(x|y)] \\ &\quad + \frac{1}{2} \mathbb{E}_{p(x,y)} \log \frac{p(x, y)}{\sqrt{p(x|y)p(y|x)}}. \end{aligned} \quad (9)$$

Theorem B.3 (Variable lower bound reconstruction). Mutual information satisfies the variational lower bound:

$$I(X; Y) \geq \mathcal{J}_{VLB} = -\frac{1}{2} [\mathbb{E} \log r(y|x) + \mathbb{E} \log r(x|y)], \quad (10)$$

where r is an arbitrary family of variational distributions.

Proof. According to the definition of conditional entropy $H(Y|X)$, we introduce the variational distribution $r(y|x)$ to approximate the real conditional distribution $p(y|x)$. It follows from Gibbs inequality that for any probability distribution $r(y|x)$ there is:

$$H(Y|X) \leq -\mathbb{E}_{p(x,y)} \log r(y|x). \quad (11)$$

Similarly, we have:

$$H(X|Y) \leq -\mathbb{E}_{p(x,y)} \log r(x|y). \quad (12)$$

Substituting these two inequalities into the decomposition of Lemma 3.2 yields:

$$\begin{aligned} I(X; Y) &\geq \frac{1}{2} (-\mathbb{E} \log r(y|x) - \mathbb{E} \log r(x|y)) \\ &\quad + \frac{1}{2} D_{KL}(p(x, y) \| \sqrt{p(x|y)p(y|x)}). \end{aligned} \quad (13)$$

Due to the non-negativity of the Kullback-Leibler divergence $D_{KL}(\cdot \| \cdot)$, a looser lower bound is obtained by discarding the end terms:

$$I(X; Y) \geq -\frac{1}{2} \mathbb{E} (\log r(y|x) + \log r(x|y)). \quad (14)$$

This is the variational lower bound $\mathcal{J}_{\text{VLB}}$ stated in the theorem.

The model can learn the representation of the molecule by maximizing the MI between the 2D and 3D. In previous approaches, they establish alignment of graph-level representations, which may result in loss of local detail information. Moreover, they lack consideration of redundant information between modalities.

3.3.2. Effective and compact representation reconstruction

Graph reconstruction on molecular space is a complex problem. It requires algorithms that can efficiently handle discretization (atoms and bonds are discrete entities) and follow the rules of chemistry. In addition, the number of possible structures of a molecule is enormous, which leads to the characterization of the large search space. Therefore, we establish generative tasks for mutual reconstruction in the representation space. It allows the encoder of each modality to understand the data from the perspective of the other modal.

We propose a method for learning cross-modal data representations by constructing coupled priors in the framework of variational generative models. We optimize and tune the respective encoders by exploiting the information of one modality as a supervised signal for the learning of the other modality. Through this cross-modal interactive learning, the encoders are guided to extract more information that is complementary to understanding and learning from the other modality. Meanwhile, we aim to reduce redundant information in the latent representations, which includes compressing the representation of each modality and reducing information that is not contributing new information or is already adequately represented by other modalities, thus improving the efficiency and generalization of the representation.

To formalize the interaction between modalities, we define a loss function that ensures each modality can reconstruct the other while maintaining consistency with the learned latent structure.

Definition 3.3 (Variational Cross-modal Reconstruction Loss). For encoder $q(z_{x_i}|h_{x_i})$ and decoder $p(h_{x_j}|z_{x_i})$, we define the loss from h_{x_i} to h_{x_j} as follows (Higgins et al., 2017):

$$\mathcal{L}_{h_{x_i} \rightarrow h_{x_j}} = -\mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p(h_{x_j}|z_{x_i})] + \beta \cdot KL(q(z_{x_i}|h_{x_i})||p_0(z_{x_i})). \quad (15)$$

where $p_0(z) = \mathcal{N}(0, I)$ is a standard Gaussian prior and β is the regularization factor. Symmetrically, we can define:

$$\mathcal{L}_{h_{x_j} \rightarrow h_{x_i}} = -\mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p(h_{x_i}|z_{x_j})] + \beta \cdot KL(q(z_{x_j}|h_{x_j})||p_0(z_{x_j})). \quad (16)$$

This loss ensures bidirectional reconstruction between modalities, promoting complementary information extraction. Building on the reconstruction loss, we introduce a coupled prior to explicitly model the interactions between latent variables of different modalities.

Definition 3.4 (Coupling Prior). Construct the following coupled prior distribution for the latent variable pair (x, y) (Jaynes, 1957).

$$p(z_{x_i}, z_{x_j}) = \frac{p(z_{x_i})p(z_{x_j}) \exp(\gamma_1 T_1(z_{x_i}, z_{x_j}) + \gamma_2 T_2(z_{x_i}, z_{x_j}))}{C}, \quad (17)$$

where $T_1(z_{x_i}, z_{x_j})$ and $T_2(z_{x_i}, z_{x_j})$ are sufficient statistics to characterize the relationship between z_{x_i} and z_{x_j} . γ_1 and γ_2 are hyperparameters that control the strength of these interactions. C is the normalization constant.

The coupled prior captures the statistical dependencies between modalities, enabling the model to learn latent representations that reflect both modality-specific and cross-modal relationships.

Lemma 3.1 (Cross-modal Variational Lower Bound). The variational posterior distribution $q(z_{x_i}, z_{x_j} | h_{x_i}, h_{x_j}) = q(z_{x_i} | h_{x_i})q(z_{x_j} | h_{x_j})$, for the joint log-likelihood $\log p(h_{x_i}, h_{x_j})$, has an evidence lower bound (ELBO)

that can be expanded as:

$$\begin{aligned} \text{ELBO} &= \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p(h_{x_i}|z_{x_i})] - \text{KL}(q(z_{x_i}|h_{x_i})||p_0(z_{x_i})) \\ &\quad + \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p(h_{x_j}|z_{x_j})] - \text{KL}(q(z_{x_j}|h_{x_j})||p_0(z_{x_j})) \\ &\quad + \mathbb{E}_{q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j})} [\gamma_1 T_1(z_{x_i}, z_{x_j}) + \gamma_2 T_2(z_{x_i}, z_{x_j}) - \log C(\gamma_1, \gamma_2)]. \end{aligned} \quad (18)$$

Proof of Lemma 3.1.

Lemma B.4 (Maximum Entropy/Minimum KL Principle Jaynes, 1957): Let $\mathcal{P}$ be a family of probability distributions for X , and P_0 a baseline distribution. The distribution $P^* \in \mathcal{P}$ minimizing $KL(P||P_0)$, under the constraints $\mathbb{E}_P[T_k(X)] = t_k$ for $k = 1, \dots, K$, is given by:

$$P^*(x) = \frac{P_0(x) \exp\left(\sum_{k=1}^K \gamma_k T_k(x)\right)}{C}, \quad (19)$$

where γ_k are Lagrange multipliers for the constraints, and $C = \mathbb{E}_{P_0}[\exp(\sum_{k=1}^K \gamma_k T_k(x))]$ is the normalization constant.

Joint log-likelihood $\log p(h_{x_i}, h_{x_j})$:

$$\begin{aligned} \log p(h_{x_i}, h_{x_j}) &= \log \iint p(h_{x_i}, h_{x_j}, z_{x_i}, z_{x_j}) dz_{x_i} dz_{x_j} \\ &= \log \iint q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j}) \frac{p(h_{x_i}, h_{x_j}, z_{x_i}, z_{x_j})}{q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j})} dz_{x_i} dz_{x_j} \\ &\geq \iint q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j}) \log \frac{p(h_{x_i}, h_{x_j}, z_{x_i}, z_{x_j})}{q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j})} dz_{x_i} dz_{x_j} \\ &= \mathbb{E}_{q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j})} [\log p(h_{x_i}, h_{x_j}, z_{x_i}, z_{x_j}) \\ &\quad - \log q(z_{x_i}|h_{x_i}) - \log q(z_{x_j}|h_{x_j})]. \end{aligned} \quad (20)$$

Substitute $p(h_{x_i}, h_{x_j}, z_{x_i}, z_{x_j})$:

$$\begin{aligned} \text{ELBO} &= \mathbb{E}_{q_i q_j} [\log p(h_{x_j}|z_{x_i}) + \log p(h_{x_i}|z_{x_j}) + \log p(z_{x_i}, z_{x_j}) \\ &\quad - \log q(z_{x_i}|h_{x_i}) - \log q(z_{x_j}|h_{x_j})] \\ &= \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p(h_{x_j}|z_{x_i})] + \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p(h_{x_i}|z_{x_j})] \\ &\quad - \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log q(z_{x_i}|h_{x_i})] - \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log q(z_{x_j}|h_{x_j})] \\ &\quad + \mathbb{E}_{q(z_{x_i}|h_{x_i})q(z_{x_j}|h_{x_j})} [\log p(z_{x_i}, z_{x_j})]. \end{aligned} \quad (21)$$

For $\mathbb{E}_{q_i q_j} [\log p(z_{x_i}, z_{x_j})]$, substitute the coupled prior of Definition 3.4:

$$p(z_{x_i}, z_{x_j}) = \frac{p_0(z_{x_i})p_0(z_{x_j}) \exp(\gamma_1 T_1(z_{x_i}, z_{x_j}) + \gamma_2 T_2(z_{x_i}, z_{x_j}))}{C(\gamma_1, \gamma_2)}, \quad (22)$$

$$\begin{aligned} \mathbb{E}_{q_i q_j} [\log p(z_{x_i}, z_{x_j})] &= \mathbb{E}_{q_i q_j} [\log p_0(z_{x_i}) + \log p_0(z_{x_j}) + \gamma_1 T_1 + \gamma_2 T_2 - \log C] \\ &= \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p_0(z_{x_i})] + \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p_0(z_{x_j})] \\ &\quad + \mathbb{E}_{q_i q_j} [\gamma_1 T_1 + \gamma_2 T_2 - \log C]. \end{aligned} \quad (23)$$

Substitute this back into the ELBO expression:

$$\begin{aligned} \text{ELBO} &= \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p(h_{x_j}|z_{x_i})] \\ &\quad - (\mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log q(z_{x_i}|h_{x_i})] - \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p_0(z_{x_i})]) \\ &\quad + \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p(h_{x_i}|z_{x_j})] \\ &\quad - (\mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log q(z_{x_j}|h_{x_j})] - \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p_0(z_{x_j})]) \\ &\quad + \mathbb{E}_{q_i q_j} [\gamma_1 T_1 + \gamma_2 T_2 - \log C] \\ &= \mathbb{E}_{q(z_{x_i}|h_{x_i})} [\log p(h_{x_j}|z_{x_i})] - \text{KL}(q(z_{x_i}|h_{x_i})||p_0(z_{x_i})) \\ &\quad + \mathbb{E}_{q(z_{x_j}|h_{x_j})} [\log p(h_{x_i}|z_{x_j})] - \text{KL}(q(z_{x_j}|h_{x_j})||p_0(z_{x_j})) \\ &\quad + \mathbb{E}_{q_i q_j} [\gamma_1 T_1(z_{x_i}, z_{x_j}) + \gamma_2 T_2(z_{x_i}, z_{x_j}) - \log C(\gamma_1, \gamma_2)]. \end{aligned} \quad (24)$$

The design of the statistics T_1 and T_2 derives from the maximum entropy principle for exponential family distributions. The exponential family is the form of distribution that maximizes entropy, provided that the expectation constraints are satisfied.

Theorem 3.2 (Generative Objective for Effective Cross-modal Representation Reconstruction). We expect to preserve the complementary information between the two modes in the potential space on one hand, and align their variational posteriors to reduce redundancy on the other hand. We take the mutual information and posterior alignment as the desired constraint, and embed the regular term through the corresponding sufficient statistics. Therefore, T_1 and T_2 can be defined as:

$$\begin{aligned} T_1(z_{x_i}, z_{x_j}) &= \log \frac{q(z_{x_i}, z_{x_j})}{q(z_{x_i})q(z_{x_j})}, \\ T_2(z_{x_i}, z_{x_j}) &= \log \frac{q(z_{x_i} | h_{x_i})}{q(z_{x_j} | h_{x_j})}. \end{aligned} \quad (25)$$

Considering the extraction of cross-modal complementary information and the reduction of redundancy, we base on maximizing the ELBO and integrating the β -weighted loss in Definition 3.3, the final optimization objective $\mathcal{L}_G$ can be defined as:

$$\begin{aligned} \mathcal{L}_G &= \frac{1}{2} (\mathcal{L}_{h_{x_i} \rightarrow h_{x_j}} + \mathcal{L}_{h_{x_j} \rightarrow h_{x_i}}) \\ &\quad - \gamma_1 I(z_{x_i}; z_{x_j}) + \gamma_2 \text{KL}(q(z_{x_i} | h_{x_i}) \| q(z_{x_j} | h_{x_j})). \end{aligned} \quad (26)$$

The constant term $\log C$ is omitted because it does not affect the optimization.

Coupled prior optimization works like capturing statistical dependencies between 2D and 3D modalities. The variational objective $\mathcal{L}_G$ is designed to eliminate redundant and useless noise while preserving the unique strengths of both views, yielding a compact yet fully informative molecular representation. The terms in Eq. (26) are intuitive. The first term is variational cross-modal reconstruction loss, which strengthens cross-modal supervision to extract relevant complementary information. The second term serves as an estimate or its proxy for the mutual information between latent variables z_{x_i} and z_{x_j} , preserving signal information. The third term reduces redundancy in the representation strategy by minimizing the difference between the two distributions.

3.3.3. Node-level contrastive objective

Contrastive self-supervised learning is considered to be an effective synergistic approach to generative methods. To mitigate the possible information loss problem in the representation space, we enhance the mutual information between the node-level representations by constructing contrastive loss functions InfoNCE (van den et al., 2018). Specifically, we define that for each molecule in each batch, the nodes of its 2D modality form a set of positive pairs with the corresponding nodes of its 3D modality. Correspondingly, for a particular 2D modality node, all other 3D modality nodes except its own are its negative pairs. The node-level contrastive loss is calculated as follows:

$$\mathcal{L}_{z_{2D}, z_{3D}} = -\frac{1}{B_n} \sum_{i=1}^{B_n} \log \frac{\exp(s(z_{2D,i}, z_{3D,i})/\tau)}{\sum_{j=1, j \neq i}^{B_n} \exp(s(z_{2D,i}, z_{3D,j})/\tau)}, \quad (27)$$

where s is the similarity function that uses the inner product as measure, B_n is the total number of atoms in the molecule within batch, and τ is the temperature parameter. Similarly, we have $\mathcal{L}_{z_{3D}, z_{2D}}$. The total loss of this part is

$$\mathcal{L}_{NCL} = \frac{1}{2} (\mathcal{L}_{z_{2D}, z_{3D}} + \mathcal{L}_{z_{3D}, z_{2D}}). \quad (28)$$

3.3.4. Training objective

The generative task, with its sufficient mutual reconstruction mechanism, and the node-level contrastive learning task, each capture different features of the molecule by targeting distinct aspects through separate learning mechanisms and objectives. By combining these two SSL

Table 1

Hyperparameter specifications for ECMMR.

Hyperparameter	Value
epochs	{50, 100}
learning rate 2D GNN	{1e-5, 1e-6}
learning rate 3D GNN	{1e-5, 1e-6}
masking ratio M	{0.15, 0.3}
β	{0.01, 0.1, 1}
γ_1	{0.01, 0.1, 1}
γ_2	{0.01, 0.1, 1}

tasks, the model's overall understanding of the molecular structure is significantly enhanced. The final loss function is defined as:

$$\mathcal{L} = \mathcal{L}_G + \mathcal{L}_{NCL}. \quad (29)$$

4. Experiments

4.1. Datasets

The PCQM4Mv2 (Hu et al., 2020) dataset was used as our pre-training dataset. This is a large-scale dataset containing about 3.38M training molecules. For downstream tasks, We select 8 datasets from MoleculeNet (Wu et al., 2018) for the classification task. The 8 datasets are divided into training, validation and test sets in a ratio of 8:1:1 according to the molecular scaffolds. The QM9 dataset (Ramakrishnan et al., 2014) contains about 134K small molecules providing quantum chemical properties such as energy and thermodynamics. We split QM9 into 110K for training, 10K for validation, and 11K for testing. The MD17 (Chmiela et al., 2017) dataset contains eight tasks and we use 1K samples for training, 1K samples for validation, and a test set consisting of 48K to 991K molecules for different tasks.

4.2. Comparison methods

To validate the effectiveness of our method, in the classification task we select 9 methods for comparison and in regression task 5 methods are compared. For GIN (Xu et al., 2018) and Schütt et al. (2017), supervised learning was performed directly using random parameter initialization. AttrMask, ContextPred (Hu et al., 2019), InfoGraph (Sun et al., 2020), MGSSL (Zhang et al., 2021) and MolCLR (Wang et al., 2022) are pretrained by designing self-supervised tasks on single modality of molecules. HGMVP, a method based on our approach, employs variational representation reconstruction to perform generative proxy tasks. 3D Infomax (Stärk et al., 2022), GraphMVP (Liu et al., 2021), MoleculesSDE (Liu et al., 2023) and MolGT (Chen et al., 2025) are pretrained on molecular multi-modal data and tested on both tasks.

4.3. Implementation details

Following the previous method's setup, we use a 5-layer GIN as the molecular 2D encoder and SchNet as the molecular 3D encoder. For a fair comparison, we eliminate the molecular fragment-enhanced hypergraph in the fine-tuning stage. This ensures that our model shares the exact same architecture as the baselines during evaluation, confirming that the observed performance gains stem solely from the effectiveness of our pre-trained representations. We conducted our experiments on a Linux server equipped with RTX 3090 GPU and Intel(R) Xeon(R) Gold 6330 CPU @ 2.00GHz. In Table 1, we illustrate the key hyperparameters used in ECMMR.

4.4. Results and analysis

Table 2 shows the results of all compared methods on the MoleculeNet dataset, from which we have the following observations. ECMMR

Table 2

The results of 8 molecular property prediction classification tasks on MoleculeNet. The best and second best ROC-AUC (%) results are highlighted in bold and underlined, respectively. We experimented with 3 seeds and calculated the mean and standard deviation of the evaluation metrics.

Methods	BBBP	Tox21	ToxCast	SIDER	ClinTox	MUV	HIV	BACE	Avg
GIN	68.1 ± 0.6	75.3 ± 0.2	62.1 ± 0.2	57.0 ± 1.3	83.7 ± 2.9	74.6 ± 2.4	75.2 ± 0.7	76.7 ± 2.5	71.6
AttrMask	65.0 ± 2.4	74.8 ± 0.2	62.9 ± 0.1	61.0 ± 0.1	87.7 ± 1.2	73.4 ± 2.0	76.8 ± 0.5	79.7 ± 0.3	72.6
ContextPred	65.7 ± 0.6	74.2 ± 0.1	62.5 ± 0.3	60.9 ± 0.6	74.0 ± 3.4	75.3 ± 1.8	77.1 ± 0.9	76.0 ± 2.1	70.7
InfoGraph	67.5 ± 0.1	73.2 ± 0.4	63.7 ± 0.5	59.9 ± 0.3	76.5 ± 1.1	74.1 ± 0.7	75.1 ± 1.0	77.8 ± 0.8	70.9
MGSSL	70.5 ± 1.2	<u>75.4 ± 0.4</u>	63.4 ± 0.3	56.4 ± 0.6	80.3 ± 1.9	72.2 ± 0.8	77.0 ± 0.3	79.7 ± 0.8	71.9
MolCLR	66.6 ± 1.9	73.0 ± 0.2	62.9 ± 0.4	57.5 ± 1.8	86.1 ± 0.9	72.5 ± 2.4	76.2 ± 1.5	71.5 ± 3.0	70.8
3D InfoMax	68.3 ± 1.1	74.5 ± 0.7	64.4 ± 0.8	60.6 ± 0.9	79.9 ± 3.5	76.2 ± 1.4	76.1 ± 1.3	79.7 ± 1.5	72.5
GraphMVP	68.5 ± 0.2	74.3 ± 0.4	63.1 ± 0.2	60.5 ± 0.3	79.1 ± 2.8	75.0 ± 1.4	76.1 ± 0.8	79.8 ± 0.7	72.1
MoleculeSDE	69.2 ± 0.4	75.3 ± 0.4	63.5 ± 0.6	60.2 ± 0.8	82.9 ± 1.2	77.2 ± 3.4	76.7 ± 0.6	80.4 ± 0.9	73.2
MolGT	<u>73.3 ± 0.5</u>	74.8 ± 0.1	<u>65.1 ± 0.4</u>	65.1 ± 0.6	<u>88.1 ± 1.4</u>	<u>78.0 ± 2.6</u>	78.4 ± 1.2	82.8 ± 0.6	<u>75.7</u>
ECMMR (ours)	75.6 ± 0.2	76.3 ± 0.8	65.4 ± 0.5	<u>63.2 ± 0.3</u>	91.3 ± 0.9	78.3 ± 1.5	<u>78.0 ± 0.5</u>	<u>81.6 ± 0.6</u>	76.2

Table 3

The results of 12 molecular property prediction regression tasks on QM9. The best and second best MAE results are highlighted in bold and underlined, respectively.

Methods	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	μ	C_v	G	H	R^2	U	U_0	$ZPVE$
SchNet	0.066	45.63	29.39	23.54	0.029	0.032	14.32	14.82	0.166	13.95	14.50	1.777
3D Infomax	0.066	42.48	26.05	21.99	0.030	0.031	13.77	13.73	0.217	12.40	12.36	1.637
GraphMVP	0.062	44.26	27.12	22.49	<u>0.027</u>	0.031	14.33	14.01	0.137	13.96	13.74	1.640
HGMVP	0.065	43.47	26.09	21.98	0.028	0.031	12.33	14.59	0.226	13.07	12.84	1.588
MoleculeSDE	<u>0.057</u>	<u>42.56</u>	<u>25.92</u>	<u>21.75</u>	0.028	<u>0.029</u>	<u>12.06</u>	11.11	0.245	<u>11.34</u>	10.98	<u>1.503</u>
ECMMR (ours)	0.056	41.04	25.33	21.69	0.026	0.027	11.84	<u>11.27</u>	<u>0.153</u>	11.19	<u>11.38</u>	1.493

Table 4

The results of 8 force prediction regression tasks on MD17. The best and second best MAE results are highlighted in bold and underlined, respectively.

Methods	Aspirin	Benzene	Ethanol	Malonaldehyde	Naphthalene	Salicylic	Toluene	Uracil
SchNet	1.391	0.359	0.458	0.832	0.589	0.803	0.533	0.784
3D Infomax	1.405	0.399	0.429	0.906	0.629	0.793	0.556	0.754
GraphMVP	1.282	0.392	<u>0.389</u>	0.853	0.575	0.744	0.479	0.778
HGMVP	1.255	0.382	0.423	0.830	<u>0.574</u>	0.741	0.476	0.741
MoleculeSDE	<u>1.211</u>	<u>0.330</u>	0.423	0.615	0.584	<u>0.728</u>	<u>0.453</u>	<u>0.727</u>
ECMMR (ours)	1.063	0.254	0.387	0.571	0.401	0.677	0.442	0.591

significantly outperforms all baseline models. Specifically, ECMMR achieves 4.6% higher average performance than the random initialization model. Our approach demonstrated superior performance across five tasks BBBP, Tox21, ToxCast, ClinTox, and MUV, with particularly outstanding results on the ClinTox task. Since toxicity is often triggered by specific functional groups, our use of the BRICS algorithm to explicitly decompose molecules allows the model to capture these critical local chemical environments more effectively than standard GNNs that treat all atoms with uniform priority. We can find that almost all multi-modal based methods have improvement. This shows that the complementary information of molecular multimodality can be utilized rationally and is task effective. ECMMR achieves better performance compared to the advanced molecular multi-modal based methods. This superiority stems from our coupled-priori optimization, which does not only utilizes both 2D and 3D features but actively filters out modality redundancy. By focusing on the consistent semantic information between topological structures and spatial conformations, ECMMR learns a more compact and noise-resistant representation.

Tables 3 and 4 show the performance results for total 20 regression tasks on QM9 and MD17 datasets, respectively. Specifically, among the 12 tasks in QM9, ECMMR achieved optimal performance in 9 tasks. Although it slightly underperformed MoleculeSDE in individual tasks such as H , it remained significantly superior to all comparison methods overall. In the molecular force prediction task for MD17, ECMMR demonstrated comprehensive superiority, achieving the lowest mean absolute error across all 8 molecular systems. This highlights its distinct advantage in capturing interatomic interactions and structure-property relationships. Our method achieves the best performance on 17 of these

20 tasks. This comprehensive performance validates the effectiveness of the ECMMR modeling strategy in molecular representation learning.

In comparison with existing methods in data space and representation space, our method demonstrates significant advantages. These existing methods usually ignore modality redundancy and the partitioning information of functional groups in molecules, resulting in their sub-optimal performance. By training the model to perform the mutual reconstruction proxy task between 2D and 3D modal information, we ensure the latent space incorporates sufficient geometric constraints. Consequently, the model gains a deeper understanding of how the spatial arrangement of functional groups determines molecular behavior. Furthermore, ECMMR's outstanding performance further validates the potential of exploring molecular representation spaces guided by chemical knowledge-based priors.

4.5. Ablation studies

4.5.1. Effect of ECMMR's loss function components

Tables 5–7 present the ablation analysis, demonstrating the effectiveness of each component within the ECMMR loss function. In classification tasks, removing $\mathcal{L}_G$ leads to a significant decline in average performance. For example, the average AUC in Table 5 drops from 76.2 to 73.3. This validates the necessity of optimizing latent space representations via $\mathcal{L}_G$. This design not only ensures bidirectional semantic alignment between 2D and 3D modalities but also enhances representational sufficiency and compactness by compressing redundant information. In regression tasks, the absence of the node-level contrastive learning term $\mathcal{L}_{NCL}$ leads to performance degradation across multiple

Table 5

Results of ablation experiments on MoleculeNet for 8 molecular property prediction classification tasks. The best and second best ROC-AUC (%) results are highlighted in bold and underlined, respectively.

Method	Classification								
	BBBP	Tox21	ToxCast	SIDER	ClinTox	MUV	HIV	BACE	Avg
GIN	68.1 ± 0.6	75.3 ± 0.2	62.1 ± 0.2	57.0 ± 1.3	83.7 ± 2.9	74.6 ± 2.4	75.2 ± 0.7	76.7 ± 2.5	71.6
ECMMR (w/o L_G)	70.5 ± 0.5	75.6 ± 0.6	64.9 ± 0.9	59.8 ± 0.7	87.6 ± 0.3	74.2 ± 1.7	75.8 ± 0.9	78.1 ± 0.6	73.3
ECMMR (w/o L_{NCL})	70.9 ± 0.7	74.6 ± 1.0	63.1 ± 0.4	60.1 ± 1.2	89.7 ± 2.4	77.9 ± 1.1	77.0 ± 2.3	79.6 ± 3.1	74.1
ECMMR (w/o MF)	72.1 ± 0.4	76.1 ± 0.9	64.2 ± 0.3	61.4 ± 1.3	89.8 ± 1.3	76.3 ± 0.9	77.4 ± 0.8	81.4 ± 1.0	74.8
ECMMR (ours)	75.6 ± 0.2	76.3 ± 0.8	65.4 ± 0.5	63.2 ± 0.3	91.3 ± 0.9	78.3 ± 1.5	78.0 ± 0.5	81.6 ± 0.6	76.2

Table 6

Ablation experiments on generative pre-training tasks. Experimental results on 12 tasks in the QM9 dataset.

Methods	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	μ	C_v	G	H	R^2	U	U_0	$ZPVE$
ours (w/o L_{NCL})	0.061	44.21	27.26	22.81	0.029	0.031	14.19	14.21	0.121	11.14	13.68	1.748
ours (w/o L_G)	0.065	41.49	25.95	22.22	0.027	0.029	12.99	12.24	0.219	12.48	12.31	1.591
ours (VRR)	0.065	43.47	25.73	21.98	0.028	0.031	12.33	14.59	0.226	13.07	12.84	1.547
ours (w/o MF)	0.058	41.89	25.81	21.75	0.027	0.029	11.89	11.96	0.189	12.01	12.88	1.672
ECMMR	0.056	41.04	25.33	21.69	0.026	0.027	11.84	11.27	0.153	11.19	11.38	1.493

Table 7

Ablation experiments on generative pre-training tasks. Experimental results on 8 tasks in the MD17 dataset.

Methods	Aspirin	Benzene	Ethanol	Malonaldehyde	Naphthalene	Salicylic	Toluene	Uracil
ours (w/o L_{NCL})	1.136	0.273	0.381	0.736	0.598	0.838	0.528	0.722
ours (w/o L_G)	1.207	0.309	0.460	0.803	0.696	0.946	0.587	0.645
ours (VRR)	1.255	0.382	0.423	0.830	0.574	0.741	0.476	0.741
ours (w/o MF)	1.109	0.331	0.413	0.677	0.523	0.795	0.475	0.629
ECMMR	1.063	0.254	0.387	0.571	0.401	0.677	0.442	0.591

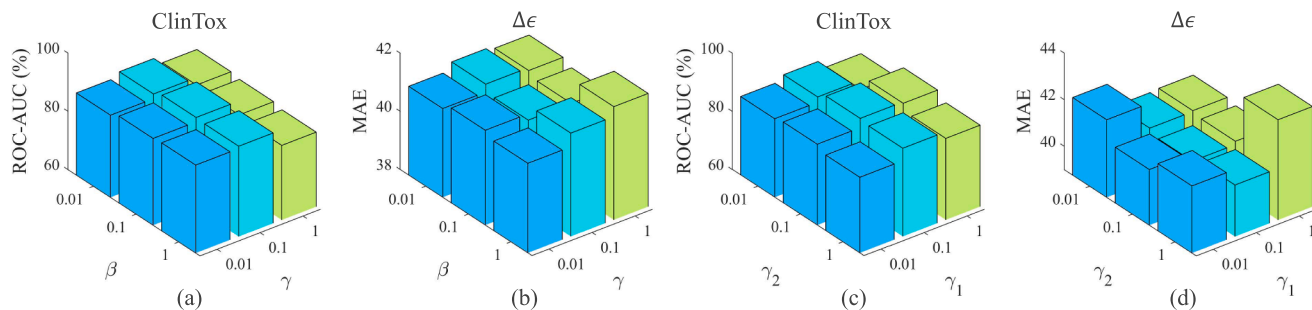


Fig. 2. Parameter sensitive analysis of ECMMR. In (a) and (b), we show the results of ablation experiments for γ with β when $\gamma_1 = \gamma_2$. And in parts (c) and (d), we analyze the ablation experiments of adjusting γ_1 and γ_2 individually while keeping β constant.

metrics. This indicates that L_{NCL} effectively estimates atomic-level mutual information, thereby preventing the loss of local molecular fine-structure information while achieving global feature alignment. Our findings reveal that combining L_G with L_{NCL} further enhances the model's ability to capture complex molecular properties. This integrated strategy not only addresses redundancy challenges across modalities but also synergizes with BRICS-based fragment-enhanced hypergraphs. The hypergraph structure enables the model to capture key functional group information within many-to-many atomic relationships, while the carefully designed loss function ensures efficient reconstruction of these higher-order structural features in the latent space. The ECMMR model successfully focuses on the importance of functional group fragments within a molecule, effectively mitigating the redundancy and information scarcity that exists among molecular views. The model achieves bi-directional optimization in the latent space through a well-designed self-supervised learning task, which significantly enhances the model's expressive power.

4.5.2. Effect of the molecular fragment-enhanced hypergraph

Tables 5–7 present the ablation study of the Molecular Fragment-Enhanced Hypergraph (MF-HG). Incorporating MF-HG further enhances

performance, underscoring the value of molecular fragments with functional groups and confirming the superiority of ECMMR. In addition, we attempted to incorporate the molecular fragment-enhanced hypergraph on each layer of the GNN. Large fluctuations in the pretraining process. Our test results reveal a significant performance degradation, even below the unpretrained baseline. In multilayer GNNs, the flow of information at each layer is critical to the final performance. It is possible that the extra information contained in the hypergraph may be beyond the processing power of the model, resulting in important features being overwhelmed by noise. We therefore suggest constructing molecular fragment enhancement in the last GNN layer.

We additionally explore the comparison with other fragmentation methods except BRICS. We developed additional molecular fragmentation methods based on our approach, randomized molecular fragmentation based on width/depth-first search. The experimental results are shown in Table 8. Specifically, we randomly select a number of fragments up to the number of atoms and use a width/depth-first search to divide the molecule into the specified number of segments, ensuring each fragment contains at least one atom. Experimental results show that molecular fragmentation using BRICS achieves the best results. The reason lies in the fact that fragments generated by BRICS possess

Table 8
Comparison of molecular fragmentation algorithms.

Methods	BACE	Clintox
ECMMR-dfs	80.6 ± 0.8	90.8 ± 0.7
ECMMR-bfs	80.7 ± 0.5	91.0 ± 1.2
ECMMR	81.6 ± 0.6	91.3 ± 0.9

Table 9
Computational cost on pretraining algorithm.

Algorithm	min/epoch
3D Infomax	10 min/epoch
GraphMVP	12 min/epoch
HGMVP	23 min/epoch
MoleculeSDE	39 min/epoch
ECMMR	30 min/epoch

explicit chemical semantics and pharmacological context, enabling them to more effectively mimic the chemical behavior of real molecules. This discovery not only demonstrates that ECMMR can effectively leverage the biochemical prior knowledge of the BRICS algorithm, but also suggests that deeply coupling classical chemical rules with modern deep learning models represents a key pathway to achieving high-quality molecular characterization.

4.5.3. Parameter sensitivity analysis

Fig. 2 presents the results of our hyperparameter sensitivity analysis for ECMMR. In classification and regression tasks, excessively high γ or improper β settings led to performance degradation. This indicates that unilateral intensification cannot take into account the complementary and consistent properties between modalities. The settings of medium strength show a greater advantage. Empirically we set them to 0.1.

4.5.4. Complexity discussion

In terms of computational efficiency, we analyzed the time complexity of the two core methods separately. The time complexity of the contrastive learning part is about $O((B \cdot N)^2 \cdot d)$, where B denotes the batch size, N is the number of atomic nodes, and d is the hidden layer dimension. The complexity of the generative approach is about $O(B^2)$. Statistically, the average number of atoms of molecules in the training set is 12.47. In practice, the training task can be efficiently accomplished using a single RTX 3090, with no significant computational overhead introduced. The specific training time comparisons of the methods pre-trained on molecular multimodal data are detailed in Table 9.

5. Conclusion

In this paper, we propose effective and compact multimodal molecular representation optimization with molecular fragments enhancement. We first introduce a molecular fragment-enhanced hypergraph that leverages the BRICS algorithm to partition molecules into functional groups, enabling the capture of many-to-many relationships and high-order chemical semantics within pharmacophores. Further, we design generative and contrastive proxy tasks for self-supervised pretraining to learn modal complementary information with reduced redundancy. In the generative task, we propose an effective and compact representation reconstruction to maintain the consistency of the 2D and 3D modalities of the molecule in the latent space while minimizing redundant information. By achieving a compact latent space through coupled prior optimization, the model can learn to filter out modality-specific noise. For the contrastive task, we implement node-level contrastive learning to reduce information loss at the graph level within the feature space. Our objective is to ensure that information from different modalities is effectively coordinated in the latent space while reducing the effects

of redundancy. Theoretical analysis and extensive experiments demonstrate the effectiveness of our approach, achieving state-of-the-art performance on 22 out of 28 downstream tasks. In future work, we plan to improve the scalability of the model by exploring more efficient graph encoders and extending the framework to encompass other modalities to further enhance its reasoning capabilities in drug discovery.

CRedit authorship contribution statement

Junyu Lin: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Project administration; **Gaokai Wang:** Formal analysis, Investigation, Writing – original draft, Writing – review & editing; **Shucheng Li:** Investigation, Writing – original draft, Writing – review & editing; **Yazhou Ren:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration; **Mei Feng:** Investigation, Writing – review & editing; **Jing He:** Writing – review & editing, Supervision; **Andre Van Zundert:** Writing – review & editing, Supervision; **Lifang He:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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